WORKSHEET 6.1

# Accumulating Change

Unit 6: Integration and Accumulation of Change

## Key ideas

- A rate multiplied by a time gives an amount. On a graph of a rate, an amount is an **area**.
- The area under a rate graph from $t = a$ to $t = b$ is the **net change** in the quantity over that interval.
- Area below the axis counts as negative: the quantity is decreasing there.
- Amount at time $b$ = amount at time $a$ + net change from $a$ to $b$.

## Worked examples

### Example 1 · A constant rate

A cyclist rides at a constant 12 km/h for 2.5 hours. How far does she travel?

**Solution.** The area of a rectangle of height 12 and width 2.5.

**30 km.**

### Example 2 · A linear rate

Water flows at $r(t) = 4 + 2t$ litres per minute for $0 \le t \le 3$. How much water flows?

**Solution.** A trapezoid with parallel sides 4 and 10 and width 3: $\frac{1}{2}(4 + 10)(3)$.

**21 litres.**

### Example 3 · A rate that changes sign

A particle has velocity $v(t) = 8 - 4t$ m/s for $0 \le t \le 3$. Find its net change in position and the total distance it travels.

**Solution.** A triangle of area 8 above the axis on $[0, 2]$ and one of area 2 below it on $[2, 3]$.

**Net change 6 m; total distance 10 m.**



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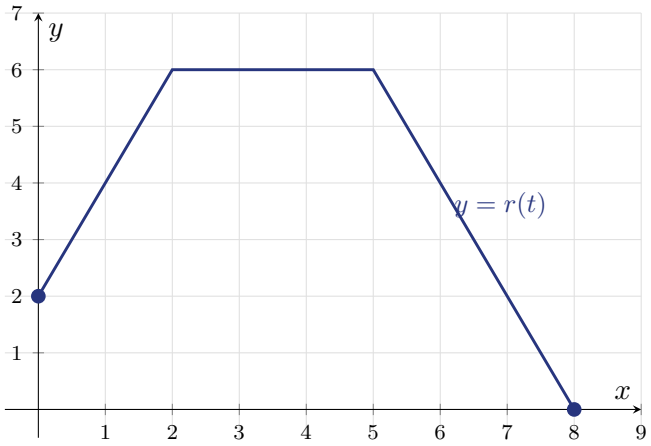
## Practice

Questions 1 to 7 use this graph. Water flows into a tank at the rate $r(t)$ litres per minute, where $t$ is in minutes.




### CORE

1. How much water flows into the tank from $t = 0$ to $t = 2$?

2. How much water flows into the tank from $t = 2$ to $t = 5$?

3. How much water flows into the tank from $t = 0$ to $t = 8$?

4. State the units of the area under the graph of $r$.

### EXAM LEVEL

5. The tank holds 10 litres at $t = 0$. How much does it hold at $t = 5$?

6. At what time have exactly 17 litres flowed into the tank?



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7. Find the average rate at which water flows into the tank over $0 \le t \le 8$.

8. A particle has velocity $v(t) = 10 - 2t$ m/s for $0 \le t \le 8$. Find its displacement and the total distance travelled.

**Question 9 · Free-response practice**

A factory makes parts at the rate $p(t)$ parts per hour during a 10-hour shift. The graph of $p$ consists of the line segments joining $(0, 40)$, $(4, 60)$, $(6, 60)$ and $(10, 20)$.

(a) How many parts are made in the first 4 hours?

(b) How many parts are made during the whole shift?

(c) Interpret $p(6) = 60$, and find the time at which the 260th part is completed.



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## Worked solutions

1. A trapezoid: $\frac{1}{2}(2+6)(2) = 8$ litres.
2. A rectangle: $6 \times 3 = 18$ litres.
3. $8 + 18 + \frac{1}{2}(6)(3) = 35$ litres.
4. (litres per minute) $\times$ (minutes) = litres.
5. $10 + 8 + 18 = 36$ litres.
6. 8 litres by $t = 2$; 9 more at 6 litres per minute takes 1.5 minutes: $t = 3.5$.
7. $\frac{35}{8} = 4.375$ litres per minute.
8. Above the axis on $[0, 5]$: 25. Below on $[5, 8]$: 9. Displacement 16 m; distance 34 m.
9. (a) $\frac{1}{2}(40+60)(4) = 200$ parts.

   (b) $200 + 60(2) + \frac{1}{2}(60+20)(4) = 200 + 120 + 160 = 480$ parts.

   (c) Six hours into the shift, parts are being made at 60 per hour. 200 parts are done by $t = 4$; 60 more at 60 per hour takes one hour: $t = 5$.



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